Solutions

Maths

Q1. Let $f:\mathbb{R}\to\mathbb{R}$ be a strictly increasing and twice differentiable function defined by $f(x)=x^3+x+2$. If $g(x)$ is the inverse of $f(x)$, then the value of $g''(4)$ is equal to

(A) $-\frac{3}{16}$ (B) $-\frac{3}{32}$ (C) $\frac{3}{32}$ (D) $-\frac{1}{16}$

Ans. B

Sol. Concept(s) : Second derivative of the inverse of a function

Key idea: Differentiate $g(f(x))=x$ twice: $g''(f(x))=-\frac{f''(x)}{(f'(x))^3}$

Given $f(x)=x^3+x+2$.
Differentiating with respect to x: $f'(x)=3x^2+1$
$f''(x)=6x$
Since $g(x)=f^{-1}(x)$, we have: $g(f(x))=x$
Differentiating both sides with respect to x: $g'(f(x))\cdot f'(x)=1 \implies g'(f(x))=\frac{1}{f'(x)}$
Differentiating again with respect to x: $g''(f(x))\cdot f'(x)=-\frac{f''(x)}{(f'(x))^2}$
$\implies g''(f(x))=-\frac{f''(x)}{(f'(x))^3}$
To find $g''(4)$, we set $f(x)=4$: $x^3+x+2=4 \implies x^3+x-2=0 \implies x=1$
Now, at $x=1$: $f'(1)=3(1)^2+1=4$
$f''(1)=6(1)=6$
Substituting these values: $g''(4)=-\frac{f''(1)}{(f'(1))^3}=-\frac{6}{4^3}=-\frac{6}{64}=-\frac{3}{32}$

Q2. Let k be the smallest positive integer for which the function $f:\mathbb{R}\to\mathbb{R}$ defined by

$$f(x)=\begin{cases}x^k\sin\left(\frac{1}{x}\right), & x\neq 0\\ 0, & x=0\end{cases}$$ is continuously differentiable at $x=0$.

An open cylindrical vessel (without a top lid) is to be constructed such that its total external surface area is fixed at $12\pi k$ square units. The maximum possible volume of this cylindrical vessel (in cubic units) is:

(A) $16\sqrt{3}\pi$ (B) $18\sqrt{3}\pi$ (C) $24\sqrt{3}\pi$ (D) $32\sqrt{3}\pi$

Ans. C

Sol. Concept(s) : Geometrical problems on maxima and minima, Continuity and differentiability of functions involving $\sin\left(\frac{1}{x}\right)$

Key idea: Continuously differentiable needs f '(x) $\to$ f '(0); the term $x^{k-2}\cos\frac{1}{x}$ forces k > 2, not just k > 1.

For $f(x)$ to be differentiable at $x=0$: $f'(0)=\lim\limits_{h\to 0}\dfrac{h^k\sin\left(\frac{1}{h}\right)-0}{h}=\lim\limits_{h\to 0}h^{k-1}\sin\left(\dfrac{1}{h}\right)$

This limit exists and equals 0 if $k-1>0 \implies k>1$.

For $x\neq 0$, the derivative is: $f'(x)=kx^{k-1}\sin\left(\dfrac{1}{x}\right)-x^{k-2}\cos\left(\dfrac{1}{x}\right)$

For $f(x)$ to be continuously differentiable at $x=0$, we require $\lim\limits_{x\to 0}f'(x)=f'(0)=0$.

This holds if $k-1>0$ and $k-2>0 \implies k>2$.

Since k is the smallest positive integer satisfying $k>2$, we have $k=3$.

Now, for the open cylinder with radius r and height h: Total Surface Area $S=\pi r^2+2\pi rh=12\pi k=36\pi. \implies h=\dfrac{36-r^2}{2r}$

The volume V is given by: $V(r)=\pi r^2h=\dfrac{\pi}{2}(36r-r^3)$

$V'(r)=\dfrac{\pi}{2}(36-3r^2)=0 \implies r=\sqrt{12}=2\sqrt{3}$

Since $V''(r)=-3\pi r<0$, $V(r)$ achieves its maximum at $r=2\sqrt{3}$.

$V_{\max}=\dfrac{\pi}{2}\left(36(2\sqrt{3})-(2\sqrt{3})^3\right)=\dfrac{\pi}{2}(72\sqrt{3}-24\sqrt{3})=24\sqrt{3}\pi$

Q3. Let $f(x)=x^3-3x^2+5x+1$. If $x=a$ is the point of inflection of the curve $y=f(x)$, then the approximate value of $f(a+0.02)$ using differentials is:

(A) 4.04 (B) 4.02 (C) 4.08 (D) 4.01

Ans. A

Sol. Concept(s) : Point of inflection and approximation using differentials

Key idea: Inflection where f " changes sign; then $f(a+\Delta x)\approx f(a)+f'(a)\Delta x$.

Differentiating $f(x)=x^3-3x^2+5x+1$ with respect to x:$f'(x)=3x^2-6x+5$

$f''(x)=6x-6=6(x-1)$

For point of inflection, $f''(x)=0 \implies x=1$.

Since $f''(x)$ changes sign at $x=1$, the point of inflection occurs at $a=1$.

Now, we need to approximate $f(1+0.02)$ using differentials:$f(a+\Delta x)\approx f(a)+f'(a)\Delta x$

Here, $a=1$ and $\Delta x=0.02$.

$f(1)=1^3-3(1)^2+5(1)+1=4$

$f'(1)=3(1)^2-6(1)+5=2$

Thus,$f(1.02)\approx 4+2(0.02)=4+0.04=4.04$

Correct Option: (A)

Q4. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function satisfying the functional equation $f(x+y) = f(x) + f(y) + 2xy \quad \forall x, y \in \mathbb{R}$ with $f'(0) = 1$.
Consider two functions $g, h : \mathbb{R} \to \mathbb{R}$ defined by

$$g(x) = \max\{f(x), f(-x), |x|\}$$

$$h(x) = \min\{f(x), f(-x), |x|\}$$

Which of the following statements is **CORRECT**?

(A) $g(x)$ is non-differentiable at 3 points in $\mathbb{R}$.

(B) $h(x)$ is non-differentiable at only 1 point in $\mathbb{R}$. (C) $h(x)$ is discontinuous at x = 0.

(D) The total number of distinct points in $\mathbb{R}$ where at least one of the functions $g(x)$ or $h(x)$ is non-differentiable is equal to 3.

Ans. D

Sol. Concept(s) : Functional equations and differentiability of max/min functions

Key idea: Get f(x) = x^2 + x, then find where the three graphs cross: g = x^2 + |x| but h switches pieces at 0,±2.

From the given functional equation, $f(x+y) = f(x) + f(y) + 2xy$
Differentiating partially with respect to x: $f'(x+y) = f'(x) + 2y$
Putting $x = 0$: $f'(y) = f'(0) + 2y = 1 + 2y \implies f'(x) = 2x + 1$
Integrating with respect to x gives $f(x) = x^2 + x + C$. Since $f(0) = 0$, $C = 0$.
Thus, $f(x) = x^2 + x$ and $f(-x) = x^2 - x$.
Now, analyzing $g(x) = \max\{x^2 + x, x^2 - x, |x|\}$:
For $x \geq 0$, $x^2 + x \geq x^2 - x$ and $x^2 + x \geq x = |x|$, so $g(x) = x^2 + x$.
For $x < 0$, $x^2 - x > x^2 + x$ and $x^2 - x - x = |x|$, so $g(x) = x^2 - x$.
Thus, $g(x) = x^2 + |x|$, which is continuous everywhere on and non-differentiable only at $x = 0$ (1 point).
Next, analyzing $h(x) = \min\{x^2 + x, x^2 - x, |x|\}$: $h(x) = \begin{cases} -x, & x < -2 \\ x^2 + x, & -2 \leq x < 0 \\ x^2 - x, & 0 \leq x \leq 2 \\ x, & x > 2 \end{cases}$
$h(x)$ is continuous everywhere on , but testing left and right derivatives shows it is non-differentiable at $x = -2, 0, 2$ (3 points).
The set of non-differentiable points for $g(x)$ is $S_1 = \{0\}$.
The set of non-differentiable points for $h(x)$ is $S_2 = \{-2, 0, 2\}$.
The set of points where at least one of $g(x)$ or $h(x)$ is non-differentiable is $S_1 \cup S_2 = \{-2, 0, 2\}$, which has exactly 3 elements.
Hence, the correct option is (D).

Q5. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 2x - \sin x, & \text{if } x \leq 0 \\ y, & \text{if } x > 0 \end{cases}$$

where for each $x > 0$, y is the unique real solution to the equation $y^3 + y = x$. Which of the following statements is/are **CORRECT** ?

(A) $f(x)$ is continuous and differentiable at $x = 0$. (B) $f'(0) = 1$. (C) $f'(2) = \frac{1}{4}$.

(D) $f''(x) > 0$ for all $x > 0$.

Ans. A, B, C

Sol. Concept(s) : Differentiability of piecewise-defined functions, Differentiation of implicit functions

Key idea: For x > 0 differentiate $y^3 + y = x$ implicitly: $f'(x) = \frac{1}{3y^2+1}$, with $y \to 0$ as $x \to 0^+$.

For $x \le 0$: $f(0) = 0$; $\lim_{x\to 0^-} f(x) = 0$

$f'(0^-) = \lim_{x\to 0^-} \frac{d}{dx}(2x - \sin x) = \lim_{x\to 0^-} (2 - \cos x) = 1.$

For $x > 0$, $y = f(x)$ satisfies $y^3 + y = x$.

As $x \to 0^+$, $y \to 0$, so $\lim_{x\to 0^+} f(x) = 0 = f(0)$. Thus, $f(x)$ is continuous at $x = 0$.

Differentiating $y^3 + y = x$ implicitly with respect to x:

$3y^2\frac{dy}{dx} + \frac{dy}{dx} = 1 \implies f'(x) = \frac{dy}{dx} = \frac{1}{3y^2+1}$

As $x \to 0^+$, $y \to 0$, hence $f'(0^+) = \frac{1}{3(0)^2+1} = 1.$

Since $f'(0^-) = f'(0^+) = 1$, $f(x)$ is differentiable at $x = 0$ and $f'(0) = 1$.

So, statements (A) and (B) are correct.

At $x = 2$: $y^3 + y = 2 \implies y = 1$.

Therefore, $f'(2) = \frac{1}{3(1)^2+1} = \frac{1}{4}$.

Differentiating $f'(x) = (3y^2 + 1)^{-1}$ with respect to x:

$f''(x) = -(3y^2+1)^{-2} \cdot 6y\frac{dy}{dx} = \frac{-6y}{(3y^2+1)^3}$

For $x > 0$, we have $y > 0$, which implies $f''(x) < 0$.

So, statement (D) is incorrect.

Q6. Consider the curves $C_1 : y = x^2 + 2$ (for $x \ge 0$) and $C_2 : y = \sqrt{x-2}$ (for $x \ge 2$). Let d_1 denote the minimum distance between the curves C_1 and C_2. Further, for a fixed point $P(0, 3)$, let d_2 denote the minimum distance from P to the curve C_1.

Which of the following statements is/are **CORRECT**?

(A) $d_1 = \frac{7}{2}$ (B) $d_1 = \frac{7\sqrt{2}}{4}$ (C) $d_2 = \frac{\sqrt{3}}{2}$ (D) $d_2 = 1$

Ans. B, C

Sol. Concept(s) : Minimum distance between two curves and from a point to a curve

Key idea: C_1 and C_2 are inverses, mirror images in y = x, so d_1 is twice the distance from C_1 to y = x.

The curves $C_1 : y = x^2 + 2$ $(x \geq 0)$ and $C_2 : y = \sqrt{x-2}$ $(x \geq 2)$ are inverse functions of each other and are symmetric about the line $y = x$.

The minimum distance between C_1 and C_2 is twice the minimum distance from C_1 to the line $y = x$.

For $C_1 : y = x^2 + 2$, the tangent parallel to $y = x$ satisfies: $\dfrac{dy}{dx} = 2x = 1 \implies x = \dfrac{1}{2}$

The corresponding point on C_1 is $Q\left(\dfrac{1}{2}, \dfrac{9}{4}\right)$.

The perpendicular distance from Q to $x - y = 0$ is: $d = \dfrac{\left|\frac{1}{2} - \frac{9}{4}\right|}{\sqrt{1^2 + (-1)^2}} = \dfrac{\frac{7}{4}}{\sqrt{2}} = \dfrac{7}{4\sqrt{2}}$

Thus, $d_1 = 2d = \dfrac{7}{2\sqrt{2}} = \dfrac{7\sqrt{2}}{4}$. Hence, (B) is correct and (A) is incorrect.

For the point $P(0, 3)$ and a point $(x, x^2 + 2)$ on C_1, the square of the distance is: $S(x) = (x - 0)^2 + (x^2 + 2 - 3)^2 = x^4 - x^2 + 1$

Differentiating with respect to x: $S'(x) = 4x^3 - 2x = 2x(2x^2 - 1)$

Setting $S'(x) = 0$ gives critical points at $x = 0$ and $x = \dfrac{1}{\sqrt{2}}$ (since $x \geq 0$).

Evaluating $S(x)$: $S(0) = 1$; $S\left(\dfrac{1}{\sqrt{2}}\right) = \left(\dfrac{1}{2}\right)^2 - \dfrac{1}{2} + 1 = \dfrac{3}{4}$

The absolute minimum of $S(x)$ occurs at $x = \dfrac{1}{\sqrt{2}}$, so $d_2 = \sqrt{\dfrac{3}{4}} = \dfrac{\sqrt{3}}{2}$.

Hence, (C) is correct and (D) is incorrect.

Q7. Let $f : (0, \infty) \to \mathbb{R}$ be a function defined by $f(x) = \begin{cases} \dfrac{\ln x}{x - 1}, & x \neq 1 \\ 1, & x = 1 \end{cases}$

For any $t \in (0, \infty) \setminus \{1\}$, let L_t denote the tangent line to the curve $y = f(x)$ at the point $(t, f(t))$. Let $g(t)$ and $h(t)$ denote the y-intercept and x-intercept of the line L_t, respectively.

Which of the following statement(s) is/are **CORRECT**?

(A) $f'(1) = -\dfrac{1}{2}$ (B) $\lim\limits_{t \to 1} g(t) = \dfrac{3}{2}$ (C) $\lim\limits_{t \to 1} h(t) = 3$ (D) $\lim\limits_{x \to 1} \dfrac{f(x) - 1 + \frac{1}{2}(x-1)}{(x-1)^2} = \dfrac{1}{3}$

Ans. A, B, C, D

Sol. Concept(s) : Equation of a tangent to a curve, L'Hospital's rule

Key idea: Near x = 1, $\dfrac{\ln x}{x - 1} = 1 - \dfrac{x-1}{2} + \dfrac{(x-1)^2}{3} - \cdots$; every limit is read from this expansion.

By definition of derivative at $x = 1$: $f'(1) = \lim\limits_{x \to 1} \dfrac{f(x) - f(1)}{x - 1} = \lim\limits_{x \to 1} \dfrac{\ln x - x + 1}{(x-1)^2}$

Applying L'Hospital's Rule ($\dfrac{0}{0}$ form): $f'(1) = \lim\limits_{x \to 1} \dfrac{\frac{1}{x} - 1}{2(x-1)} = \lim\limits_{x \to 1} \dfrac{1 - x}{2x(x-1)} = -\dfrac{1}{2}$

Hence, option (A) is correct.

The equation of tangent L_t at $(t, f(t))$ is given by: $y - f(t) = f'(t)(x - t)$

The y-intercept $g(t)$ is obtained by setting $x = 0$: $g(t) = f(t) - tf'(t)$

Since $f(t) = \dfrac{\ln t}{t - 1}$ and $f'(t) = \dfrac{t - 1 - t\ln t}{t(t-1)^2}$, we have:

$g(t) = \dfrac{\ln t}{t-1} - \dfrac{t-1-t\ln t}{(t-1)^2} = \dfrac{(2t-1)\ln t - t + 1}{(t-1)^2}$

Taking limit as $t \to 1$ ($\dfrac{0}{0}$ form) and using L'Hospital's Rule twice:

$\lim\limits_{t\to 1} g(t) = \lim\limits_{t\to 1} \dfrac{2\ln t + \frac{2t-1}{t} - 1}{2(t-1)} = \lim\limits_{t\to 1} \dfrac{2\ln t + 1 - \frac{1}{t}}{2(t-1)} = \lim\limits_{t\to 1} \dfrac{\frac{2}{t} + \frac{1}{t^2}}{2} = \dfrac{3}{2}$

Hence, option (B) is correct.

The x-intercept $h(t)$ is obtained by setting $y = 0$: $h(t) = t - \dfrac{f(t)}{f'(t)}$

Taking limit as $t \to 1$: $\lim\limits_{t\to 1} h(t) = 1 - \dfrac{\lim_{t\to 1} f(t)}{\lim_{t\to 1} f'(t)} = 1 - \dfrac{1}{-1/2} = 3$

Hence, option (C) is correct.

For option (D): $\lim\limits_{x\to 1} \dfrac{\frac{\ln x}{x-1} - 1 + \frac{1}{2}(x-1)}{(x-1)^2} = \lim\limits_{x\to 1} \dfrac{\ln x - (x-1) + \frac{1}{2}(x-1)^2}{(x-1)^3}$

Applying L'Hospital's Rule: $= \lim\limits_{x\to 1} \dfrac{\frac{1}{x} - 1 + (x-1)}{3(x-1)^2} = \lim\limits_{x\to 1} \dfrac{(x-1)^2}{3x(x-1)^2} = \dfrac{1}{3}$

Hence, option (D) is correct.

Q8. Let $f : (0,\infty) \to \mathbb{R}$ and $g : (0,\infty) \to \mathbb{R}$ be functions defined as $f(x) = x^x$ and $g(x) = x^{1/x}$, respectively. Which of the following statements is/are **CORRECT**?

(A) The limit $\lim\limits_{x\to 0^+} f(x)$ is an indeterminate form of type 0^0 and its value is 0.

(B) Exactly one tangent can be drawn to the curve $y = f(x)$ passing through the origin $(0,0)$.

(C) The function $g(x)$ attains its local maximum at $x = \dfrac{1}{e}$.

(D) The line $y = x$ is a common tangent to both curves $y = f(x)$ and $y = g(x)$ at the point $(1,1)$ and passes through the origin.

Ans. B, D

Sol. Concept(s) : Indeterminate forms 0^0 and ∞^0 via logarithms, Tangent drawn from an external point

Key idea: A tangent through the origin needs a(1+ ln a) = 1; monotonicity of $a(1+\ln a) - 1$ shows a = 1 is its only root.

For $f(x) = x^x$: is of 0^0 form.

Let $L = \lim\limits_{x\to 0^+} x^x \implies \ln L = \lim\limits_{x\to 0^+} x\ln x = \lim\limits_{x\to 0^+} \dfrac{\ln x}{1/x} = \lim\limits_{x\to 0^+} \dfrac{1/x}{-1/x^2} = 0 \implies L = e^0 = 1.$

Hence, statement (A) is incorrect.

For statement (B): Equation of tangent to $y = f(x)$ at (a, a^a) is $y - a^a = a^a(1+\ln a)(x-a)$.

If it passes through $(0,0)$, then $-a^a = -a^{a+1}(1+\ln a) \implies a(1+\ln a) = 1$.

Let $h(a) = a(1+\ln a) - 1$ for $a > 0$.

$h'(a) = 2 + \ln a$, which is negative for $a < e^{-2}$ and positive for $a > e^{-2}$.

Since $h(e^{-2}) = -e^{-2} - 1 < 0$ and $h(1) = 0$, $a = 1$ is the unique solution in $(0,\infty)$.

Thus, exactly one tangent can be drawn from the origin.

Hence, statement (B) is correct.

For $g(x) = x^{1/x}$: $g'(x) = x^{1/x}\left(\dfrac{1-\ln x}{x^2}\right)$.

$g'(x) = 0 \implies x = e$. So local maximum occurs at $x = e$, not $x = 1/e$.

Hence, statement (C) is incorrect.

For statement (D): At $x = 1$, $f(1) = 1$ and $f'(1) = 1$.

Tangent to $y = f(x)$ at $(1,1)$ is $y - 1 = 1(x-1) \implies y = x$.

Also $g(1) = 1$ and $g'(1) = 1$.

Tangent to $y = g(x)$ at $(1,1)$ is $y - 1 = 1(x-1) \implies y = x$.

This line passes through $(0,0)$.

Hence, statement (D) is correct.

Paragraph

Let $f : \mathbb{R} \to \mathbb{R}$ be a twice differentiable function satisfying $f(0) = 0$, $f'(0) = 1$, and $f''(x) + f(x)(f'(x))^2 = 0$ for all $x \in \mathbb{R}$

Consider the functions $g : \mathbb{R} \to \mathbb{R}$, $h : \mathbb{R} \to \mathbb{R}$, and $k : (0,\infty) \to \mathbb{R}$ defined by $g(x) = f(x) - x$, $h(x) = f(x)e^{\frac{x^2}{2}}$, and $k(x) = \dfrac{f(x)}{x}$

Q9. Which of the following statements is **CORRECT** regarding the monotonic behavior of the functions $f(x)$, $g(x)$, and $h(x)$ on $\mathbb{R}$?

(A) $f(x)$ is strictly increasing on $\mathbb{R}$, $g(x)$ is strictly increasing on $\mathbb{R}$, and $h(x)$ is strictly decreasing on $\mathbb{R}$.

(B) $f(x)$ is strictly increasing on $\mathbb{R}$, $g(x)$ is strictly decreasing on $\mathbb{R}$, and $h(x)$ is strictly increasing on $\mathbb{R}$.

(C) $f(x)$ is strictly decreasing on $\mathbb{R}$, $g(x)$ is strictly decreasing on $\mathbb{R}$, and $h(x)$ is strictly increasing on $\mathbb{R}$.

(D) $f(x)$ is strictly increasing on $\mathbb{R}$, $g(x)$ is strictly decreasing on $\mathbb{R}$, and $h(x)$ is strictly decreasing on $\mathbb{R}$.

Ans. B

Sol. Concept(s) : Continuity and differentiability of functions, Monotonic functions

Key idea: Divide by f' : $\dfrac{f''}{f'} = -ff'$ integrates to $f'(x) = e^{-\frac{(f(x))^2}{2}}$.

Given differential equation:

$$\frac{f''(x)}{f'(x)} = -f(x)f'(x)$$

Integrating both sides with respect to x:

$$\ln(f'(x)) = -\frac{(f(x))^2}{2} + C$$

Using $f(0) = 0$ and $f'(0) = 1$, we get $C = 0$.

$f'(x) = e^{-\frac{(f(x))^2}{2}}$

1. Since $e^t > 0$ for all $t \in \mathbb{R}$, $f'(x) > 0$ for all $x \in \mathbb{R}$. Thus, $f(x)$ is **strictly increasing** on $\mathbb{R}$.

2. For $g(x) = f(x) - x$: $g'(x) = f'(x) - 1 = e^{-\frac{(f(x))^2}{2}} - 1$ Since $(f(x))^2 \geq 0$, $e^{-\frac{(f(x))^2}{2}} \leq 1$, with equality if and only if $f(x) = 0 \iff x = 0$.

3. Hence $g'(x) < 0$ for all $x \neq 0$ and $g'(0) = 0$, implying $g(x)$ is **strictly decreasing** on $\mathbb{R}$.

4. For $h(x) = f(x)e^{\frac{x^2}{2}}$:

$$h'(x) = e^{\frac{x^2}{2}}\left(f'(x) + xf(x)\right)$$
$$= e^{\frac{x^2}{2}}\left(e^{-\frac{(f(x))^2}{2}} + xf(x)\right)$$

5. Since $f(0) = 0$ and $f(x)$ is strictly increasing:

- For $x > 0$, $f(x) > 0 \implies xf(x) > 0$.
- For $x < 0$, $f(x) < 0 \implies xf(x) > 0$.

Thus $xf(x) \geq 0$ for all $x \in \mathbb{R}$.

Since $e^{-\frac{(f(x))^2}{2}} > 0$, $h'(x) > 0$ for all $x \in \mathbb{R}$

Hence $h(x)$ is **strictly increasing** on $\mathbb{R}$.

Q10. Consider the following statements:

I. The function $k(x) = \dfrac{f(x)}{x}$ is strictly decreasing on $(0, \infty)$.

II. The equation $f(x) = cx$ has a unique real solution for every $c \in (0, 1)$.

III. $f''(x) > 0$ for all $x \in (0, \infty)$.

IV. $\lim\limits_{x \to \infty} f(x) = \dfrac{\pi}{2}$.

Which of the following options correctly identifies the truth values of the above statements?

(A) I is True, II is False, III is False, IV is False

(B) I is False, II is True, III is True, IV is False

(C) I is True, II is False, III is False, IV is True

(D) I is False, II is False, III is True, IV is True

Ans. A

Sol. Concept(s) : Continuity and differentiability of functions, Monotonic functions

Key idea: $\displaystyle\int_0^{f(x)} e^{\frac{u^2}{2}}\,du = x$ shows f $\to \infty$ but slower than any line, so $\dfrac{f(x)}{x}$ falls from 1 to 0.

1. Statement I: For $k(x) = \dfrac{f(x)}{x}$ on $(0, \infty)$:

$k'(x) = \dfrac{xf'(x) - f(x)}{x^2}$

Let $N(x) = xf'(x) - f(x)$.

Then $N'(x) = xf''(x) = -xf(x)e^{-\frac{(f(x))^2}{2}}$.

For $x > 0$, $f(x) > 0 \implies N'(x) < 0$.

Since $N(0) = 0$, $N(x) < 0$ for all $x > 0$.

Hence $k'(x) < 0$ for $x > 0$, so $k(x)$ is strictly decreasing on $(0, \infty)$. (**True**)

2. Statement II: Let $M(x) = f(x) - cx$.

$M'(x) = e^{-\frac{(f(x))^2}{2}} - c$.

Critical points occur when $(f(x))^2 = -2\ln c > 0$.

Since f is a bijection, $M'(x) = 0$ at two distinct points $x_1 < 0 < x_2$.

Since $M(0) = 0$ and $M'(0) = 1 - c > 0$, $M(x_1) < 0$ and $M(x_2) > 0$.

As $x \to \pm\infty$, $M(x) \to \mp\infty$.

By IVT, $f(x) = cx$ has **three** distinct real roots. (**False**)

3. Statement III: $f''(x) = -f(x)e^{-\frac{(f(x))^2}{2}} < 0$ for all $x > 0$. (**False**)

4. Statement IV: From $e^{\frac{(f(x))^2}{2}} f'(x) = 1$, integrating gives $\int_0^{f(x)} e^{\frac{u^2}{2}} du = x$.

As $x \to \infty$, $f(x) \to \infty$. (**False**)

Q1. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = x^3 - 3x + k$, where $k \in \mathbb{Z}$. If the equation $f(x) = 0$ has a root in the open interval $(-2, -1)$, then the number of possible integral values of k is equal to _____.

Ans. 3

Sol. Concept(s) : Monotonicity in an interval, Intermediate value theorem

Key idea: f is strictly increasing on [–2,–1], so a root lies inside exactly when f(–2)f(–1) < 0.

Differentiating $f(x)$ with respect to x, we get

$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$

For $x \in (-2, -1)$, $x^2 > 1$, which implies $f'(x) > 0$.

Thus, $f(x)$ is strictly increasing on the interval $[-2, -1]$.

Since $f(x)$ is continuous and strictly increasing on $[-2, -1]$, by the Intermediate Value Theorem, the equation $f(x) = 0$ has a root in $(-2, -1)$ if and only if

$f(-2) \cdot f(-1) < 0$

Calculating the values:

$f(-2) = (-2)^3 - 3(-2) + k = k - 2$

$f(-1) = (-1)^3 - 3(-1) + k = k + 2$

Therefore,

$(k - 2)(k + 2) < 0 \implies -2 < k < 2$

Since k must be an integer, the possible values of k are $-1, 0$, and 1.

Hence, the total number of possible integral values of k is 3.

Q2. Let $y = f(x)$ be a twice differentiable function defined implicitly for $y > 0$ by the determinant equation $\begin{vmatrix} y & \sin x & 1 \\ y^2 & \cos x & x \\ y^3 & 1 & 0 \end{vmatrix} = 0$ such that $f(0) = 1$. If $f'(0)$ and $f''(0)$ denote the first and second derivatives of $f(x)$ at $x = 0$ respectively, then the value of $f''(0) - 2f'(0)$ is equal to _____.

Ans. 3

Sol. Concept(s) : Differentiation of determinants and implicit functions

Key idea: Expand the determinant first: it becomes the plain implicit equation $y - x - y^2\cos x + xy^2\sin x = 0$.

Let the given determinant be $\Delta(x) = |C_1 \quad C_2 \quad C_3| = 0$, where

$$C_1 = \begin{pmatrix} y \\ y^2 \\ y^3 \end{pmatrix}, \quad C_2 = \begin{pmatrix} \sin x \\ \cos x \\ 1 \end{pmatrix}, \quad C_3 = \begin{pmatrix} 1 \\ x \\ 0 \end{pmatrix}$$

At $x = 0$ and $y = 1$:

$$C_1(0) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad C_2(0) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad C_3(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Differentiating $\Delta(x) = 0$ with respect to x gives:

$$\frac{d\Delta}{dx} = |C_1' \quad C_2 \quad C_3| + |C_1 \quad C_2' \quad C_3| + |C_1 \quad C_2 \quad C_3'| = 0$$

At $x = 0$, $y = 1$:

$$C_1' = y'(0)\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad C_2' = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad C_3' = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Evaluating the three determinants at $x = 0$:

1. $|C_1' \quad C_2 \quad C_3| = \begin{vmatrix} y' & 0 & 1 \\ 2y' & 1 & 0 \\ 3y' & 1 & 0 \end{vmatrix} = -y'(0)$
2. $|C_1 \quad C_2' \quad C_3| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{vmatrix} = 0$
3. $|C_1 \quad C_2 \quad C_3'| = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -1$

Thus, $-y'(0) + 0 - 1 = 0 \implies f'(0) = -1$.

Differentiating $\dfrac{d\Delta}{dx} = 0$ again with respect to x:

$$\frac{d^2\Delta}{dx^2} = |C_1'' \quad C_2 \quad C_3| + 2|C_1' \quad C_2' \quad C_3| + 2|C_1' \quad C_2 \quad C_3'|$$
$$+|C_1 \quad C_2'' \quad C_3| + 2|C_1 \quad C_2' \quad C_3'| + |C_1 \quad C_2 \quad C_3''| = 0$$

At $x = 0$, $y = 1$, $y' = -1$:

$$C_1'' = y''(0)\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + (y'(0))^2\begin{pmatrix} 0 \\ 2 \\ 6 \end{pmatrix} = \begin{pmatrix} y'' \\ 2y'' + 2 \\ 3y'' + 6 \end{pmatrix}$$

$$C_2'' = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$$

$$C_3'' = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Evaluating each term at $x = 0$:

1. $|C_1'' \quad C_2 \quad C_3| = -y''(0) - 4$
2. $|C_1' \quad C_2' \quad C_3| = 0$
3. $|C_1' \quad C_2 \quad C_3'| = 1$

4. $|C_1 \quad C_2'' \quad C_3| = 1$

5. $|C_1 \quad C_2' \quad C_3'| = 1$

6. $|C_1 \quad C_2 \quad C_3''| = 0$

Summing these terms:

$(-y''(0) - 4) + 2(0) + 2(1) + 1 + 2(1) + 0 = 0$

$\implies -y''(0) + 1 = 0 \implies f''(0) = 1$

Hence, $f''(0) - 2f'(0) = 1 - 2(-1) = 3$.

Q3. Let N be the total number of distinct common tangents to the parabola $y = x^2$ and the curve $y = x^3 - 3x + 1$. Find the value of N _____.

Ans. 2

Sol. Concept(s) : Common tangents between two curves and root counting via monotonicity and IVT

Key idea: Match slope and intercept of y = 2tx – t^2 with the cubic's tangent at x = s, then count real roots of the quartic in s.

Let $y = 2tx - t^2$ be a tangent to $y = x^2$ at the point (t, t^2).

If this line is also tangent to $y = x^3 - 3x + 1$ at $x = s$, then:

1) Slope condition: $2t = \dfrac{d}{dx}(x^3 - 3x + 1)\Big|_{x=s} = 3s^2 - 3 \implies t = \dfrac{3}{2}(s^2 - 1)$

2) y-intercept condition: $-t^2 = (s^3 - 3s + 1) - s(3s^2 - 3) = -2s^3 + 1$

Substituting $t = \dfrac{3}{2}(s^2 - 1)$ into $t^2 = 2s^3 - 1$ gives:

$\dfrac{9}{4}(s^2 - 1)^2 = 2s^3 - 1 \implies 9s^4 - 8s^3 - 18s^2 + 13 = 0$

Let $P(s) = 9s^4 - 8s^3 - 18s^2 + 13$.

To find the number of real roots of $P(s) = 0$, consider its derivative:

$P'(s) = 36s^3 - 24s^2 - 36s = 12s(3s^2 - 2s - 3)$

The critical points of $P(s)$ are $s_1 = \dfrac{1 - \sqrt{10}}{3} \approx -0.72$, $s_2 = 0$, and $s_3 = \dfrac{1 + \sqrt{10}}{3} \approx 1.39$.

Evaluating $P(s)$ using $3s^2 = 2s + 3$ at critical points:

$P(0) = 13 > 0$

$P(s_1) = 4(1 + s_1^2 - 2s_1^3) > 0$ (since $s_1 < 0$)

$P(s_3) = 4(1 + s_3^2 - 2s_3^3) < 0$ (since $s_3 > 1.38$)

Also, $\lim\limits_{s \to \pm\infty} P(s) = +\infty$.

By IVT and monotonicity:

$P(s) > 0$ for all $s \in (-\infty, s_3)$, so there are no real roots in $(-\infty, 0]$.

$P(s) = 0$ has exactly 1 real root in $(0, s_3)$.

$P(s) = 0$ has exactly 1 real root in (s_3, ∞).

Hence, $P(s) = 0$ has exactly 2 distinct real roots, each yielding a unique common tangent line.

Thus, $N = 2$.

Q4. Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = [x] + \left[x + \dfrac{1}{2}\right]$ where $[\cdot]$ denotes the greatest integer function.

Consider the function $g : (-2, 2) \to \mathbb{R}$ defined by

$$g(x) = \lim_{n\to\infty} \frac{x^{2n} f(x) + x + 2}{x^{2n} + 1}$$

If k is the total number of points in $(-2, 2)$ at which $g(x)$ is discontinuous, then find the value of k _____.

Ans. 4

Sol. Concept(s) : Discontinuities of functions defined by limits and the greatest integer function

Key idea: Hermite's identity gives f(x) = [2x]; the limit picks x + 2 for |x| < 1 and [2x] for |x| > 1.

Using the identity $[x] + \left[x + \frac{1}{2}\right] = [2x]$, we get $f(x) = [2x]$.

Now, evaluating the limit for $g(x)$:

For $|x| < 1$, i.e., $x \in (-1, 1)$: $\lim_{n\to\infty} x^{2n} = 0 \implies g(x) = x + 2$.

For $|x| > 1$, i.e., $x \in (-2, -1) \cup (1, 2)$: $\lim_{n\to\infty} x^{2n} = \infty \implies g(x) = f(x) = [2x]$.

For $x = 1$: $g(1) = \frac{f(1) + 1 + 2}{2} = \frac{[2] + 3}{2} = \frac{5}{2}$.

For $x = -1$: $g(-1) = \frac{f(-1) - 1 + 2}{2} = \frac{[-2] + 1}{2} = -\frac{1}{2}$.

Analyzing continuity of $g(x)$ in $(-2, 2)$:

1. In $(-1, 1)$: $g(x) = x + 2$ is continuous everywhere.

2. At $x = -1$: $\lim_{x\to-1^-} g(x) = \lim_{x\to-1^-} [2x] = -3$ $\lim_{x\to-1^+} g(x) = \lim_{x\to-1^+} (x + 2) = 1$

Since $\lim_{x\to-1^-} g(x) \neq \lim_{x\to-1^+} g(x)$, $x = -1$ is a point of discontinuity.

3. At $x = 1$: $\lim_{x\to1^-} g(x) = \lim_{x\to1^-} (x + 2) = 3$ $\lim_{x\to1^+} g(x) = \lim_{x\to1^+} [2x] = 2$

Since $\lim_{x\to1^-} g(x) \neq \lim_{x\to1^+} g(x)$, $x = 1$ is a point of discontinuity.

4. In $(-2, -1)$: $g(x) = [2x]$.

Since $2x \in (-4, -2)$, the integer jump occurs at $2x = -3 \implies x = -\frac{3}{2}$.

At $x = -\frac{3}{2}$, $\lim_{x\to(-3/2)^-} [2x] = -4$ and $\lim_{x\to(-3/2)^+} [2x] = -3$.

Thus, $x = -\frac{3}{2}$ is a point of discontinuity.

5. In $(1, 2)$: $g(x) = [2x]$. Since $2x \in (2, 4)$, the integer jump occurs at $2x = 3 \implies x = \frac{3}{2}$.

At $x = \frac{3}{2}$, $\lim_{x\to(3/2)^-} [2x] = 2$ and $\lim_{x\to(3/2)^+} [2x] = 3$.

Thus, $x = \frac{3}{2}$ is a point of discontinuity.

The points of discontinuity of $g(x)$ in $(-2, 2)$ are $x = -\frac{3}{2}, -1, 1, \frac{3}{2}$.

Hence, the total number of points of discontinuity is $k = 4$.

Q1. Let $f:[0,3]\to\mathbb{R}$ be a function defined by $f(x)=x^3-3x^2+4x+1$. Let $c\in(0,3)$ be the value obtained by applying Lagrange's Mean Value Theorem to $f(x)$ on the interval $[0,3]$.

Consider two curves $C_1: y=f(x)$ and $C_2: y=\dfrac{a}{x+1}+b$, where $a,b\in\mathbb{R}$ with $a>0$. If the curves C_1 and C_2 intersect orthogonally at $x=c$, then the value of a is equal to _____.

Ans. 2.25

Sol. Concept(s) : LMVT and orthogonal intersection of curves

Key idea: Orthogonal intersection means the product of the two tangent slopes at x = c is –1.

By Lagrange's Mean Value Theorem for $f(x)$ on $[0,3]$, there exists $c\in(0,3)$ such that: $f'(c)=\dfrac{f(3)-f(0)}{3-0}$

Given $f(x)=x^3-3x^2+4x+1$: $f(0)=1,\quad f(3)=27-27+12+1=13$
$f'(x)=3x^2-6x+4$

Thus, $f'(c)=\dfrac{13-1}{3}=4\implies 3c^2-6c+4=4\implies 3c(c-2)=0$

Since $c\in(0,3)$, we get $c=2$.

The slope of the tangent to C_1 at $x=2$ is $m_1=f'(2)=4$.

Let $g(x)=\dfrac{a}{x+1}+b$. The slope of the tangent to C_2 at $x=2$ is: $m_2=g'(2)=-\dfrac{a}{(x+1)^2}\Big|_{x=2}=-\dfrac{a}{9}$

Since C_1 and C_2 intersect orthogonally at $x=2$: $m_1\cdot m_2=-1\implies 4\left(-\dfrac{a}{9}\right)=-1\implies a=\dfrac{9}{4}=2.25$

Q2. Let $x(t)=t^t$ and $y(t)=t^{t^2}$ for $t>0$. If $\dfrac{dy}{dx}$ denotes the derivative of y with respect to x, evaluate the value of the expression $\dfrac{x'(t)}{x(t)}\left(2t-\dfrac{x(t)}{y(t)}\dfrac{dy}{dx}\right)$ at $t=343$, where $x'(t)=\dfrac{dx}{dt}$.

Ans. 343

Sol. Concept(s) : Logarithmic and parametric differentiation

Key idea: Work with logarithmic derivatives: $\dfrac{x}{y}\dfrac{dy}{dx}=\dfrac{y'/y}{x'/x}$, and the expression collapses to t.

Given $x(t)=t^t$ and $y(t)=t^{t^2}$.

For x: $\ln x=t\ln t$
$\dfrac{1}{x(t)}\dfrac{dx}{dt}=1+\ln t\implies\dfrac{x'(t)}{x(t)}=1+\ln t$

For $y(t)$: $\ln y=t^2\ln t$
$\dfrac{1}{y(t)}\dfrac{dy}{dt}=2t\ln t+t^2\cdot\dfrac{1}{t}=t(1+2\ln t)\implies\dfrac{y'(t)}{y(t)}=t(1+2\ln t)\dfrac{dy}{dx}=\dfrac{y'(t)}{x'(t)}$

Thus,
$\dfrac{x(t)}{y(t)}\dfrac{dy}{dx}=\dfrac{x(t)}{y(t)}\cdot\dfrac{y'(t)}{x'(t)}=\dfrac{y'(t)/y(t)}{x'(t)/x(t)}=\dfrac{t(1+2\ln t)}{1+\ln t}$

Substituting these into the given expression $E(t)$:

$E(t) = \dfrac{x'(t)}{x(t)}\left(2t - \dfrac{x(t)}{y(t)}\dfrac{dy}{dx}\right) = (1+\ln t)\left(2t - \dfrac{t(1+2\ln t)}{1+\ln t}\right)$

$E(t) = 2t(1+\ln t) - t(1+2\ln t) = 2t + 2t\ln t - t - 2t\ln t = t$

Evaluating at $t = 343$: $E(343) = 343$

Q3. Let $f(x) = \begin{vmatrix} e^x & e^{2x} & e^{3x} \\ e^x & 2e^{2x} & 3e^{3x} \\ e^x & 4e^{2x} & 9e^{3x} \end{vmatrix}$ and $g(x) = e^x$. If $y = f(x)$, then the value of $\left.\dfrac{d^3y}{d(g(x))^3}\right|_{x=0}$ is equal to ____.

Ans. 240

Sol. Concept(s) : Determinants and differentiation of one function with respect to another

Key idea: Take e^x, e^{2x}, e^{3x} out of the columns, then write y as a function of $u = e^x$.

$f(x) = e^x \cdot e^{2x} \cdot e^{3x}\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix}$

Thus, $f(x) = 2e^{6x}$

Let $u = g(x) = e^x$.

Expressing $y = f(x)$ in terms of u: $y = 2(e^x)^6 = 2u^6$

Now, differentiating y successively with respect to u:

$\dfrac{dy}{du} = 12u^5$; $\dfrac{d^2y}{du^2} = 60u^4$; $\dfrac{d^3y}{du^3} = 240u^3$

At $x = 0$, $u = g(0) = e^0 = 1$.

Therefore, $\left.\dfrac{d^3y}{d(g(x))^3}\right|_{x=0} = 240(1)^3 = 240$